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-- ==========================================
-- Titanic数据SQL探索分析 - 完整查询集
-- 分析工具：SQLite
-- 数据来源：Kaggle Titanic数据集（train.csv）
-- 记录数：891
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-- 第一部分：数据验证与概况
-- ==========================================

-- 1.1 验证数据导入 - 查看表结构
PRAGMA table_info(passengers);

-- 1.2 查看总行数
SELECT COUNT(*) AS total_passengers FROM passengers;

-- 1.3 查看前5行数据
SELECT * FROM passengers LIMIT 5;

-- 1.4 数据完整性统计（各字段缺失情况）
SELECT
    COUNT(*) AS 总人数,
    COUNT(PassengerId) AS PassengerId非空,
    COUNT(Survived) AS Survived非空,
    COUNT(Pclass) AS Pclass非空,
    COUNT(Name) AS Name非空,
    COUNT(Sex) AS Sex非空,
    COUNT(Age) AS Age非空,
    COUNT(SibSp) AS SibSp非空,
    COUNT(Parch) AS Parch非空,
    COUNT(Ticket) AS Ticket非空,
    COUNT(Fare) AS Fare非空,
    COUNT(Cabin) AS Cabin非空,
    COUNT(Embarked) AS Embarked非空
FROM passengers;

-- 1.5 关键字段缺失率详细统计
SELECT
    'Age' AS 字段,
    COUNT(*) AS 总行数,
    COUNT(Age) AS 非空值,
    COUNT(*) - COUNT(Age) AS 缺失值,
    ROUND(COUNT(Age) * 100.0 / COUNT(*), 2) AS 完整率百分比
FROM passengers
UNION ALL
SELECT
    'Cabin',
    COUNT(*),
    COUNT(Cabin),
    COUNT(*) - COUNT(Cabin),
    ROUND(COUNT(Cabin) * 100.0 / COUNT(*), 2)
FROM passengers
UNION ALL
SELECT
    'Embarked',
    COUNT(*),
    COUNT(Embarked),
    COUNT(*) - COUNT(Embarked),
    ROUND(COUNT(Embarked) * 100.0 / COUNT(*), 2)
FROM passengers;


-- ==========================================
-- 第二部分：基础分布统计
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-- 2.1 生还情况分布
SELECT
    Survived,
    COUNT(*) AS 人数,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM passengers), 2) AS 占比
FROM passengers
GROUP BY Survived;

-- 2.2 性别分布
SELECT
    Sex,
    COUNT(*) AS 人数,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM passengers), 2) AS 占比
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FROM passengers
GROUP BY Sex;

-- 2.3 舱位分布
SELECT
    Pclass,
    COUNT(*) AS 人数,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM passengers), 2) AS 占比
FROM passengers
GROUP BY Pclass
ORDER BY Pclass;

-- 2.4 登船港口分布
SELECT
    Embarked,
    COUNT(*) AS 人数,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM passengers), 2) AS 占比
FROM passengers
WHERE Embarked IS NOT NULL
GROUP BY Embarked;


-- ============================================
-- 第三部分：单变量分析 - 性别
-- ============================================

-- 3.1 性别对生还率的影响
SELECT
    CASE WHEN Sex = 'male' THEN '男性' ELSE '女性' END AS 性别,
    COUNT(*) AS 总人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
GROUP BY Sex;


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-- 第四部分：单变量分析 - 舱位
-- ============================================

-- 4.1 舱位等级对生还率的影响
SELECT
    Pclass AS 舱位等级,
    COUNT(*) AS 总人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
GROUP BY Pclass
ORDER BY Pclass;


-- ============================================
-- 第五部分：单变量分析 - 年龄
-- ============================================

-- 5.1 年龄缺失情况统计
SELECT
    COUNT(*) AS 总人数,
    COUNT(Age) AS 有年龄记录,
    COUNT(*) - COUNT(Age) AS 年龄缺失,
    ROUND(COUNT(Age) * 100.0 / COUNT(*), 2) AS 完整率百分比
FROM passengers;

-- 5.2 年龄段生还率分析
SELECT
    CASE
        WHEN Age IS NULL THEN '未知'
        WHEN Age <= 12 THEN '儿童(0-12)'
        WHEN Age <= 18 THEN '青少年(13-18)'
        WHEN Age <= 35 THEN '青年(19-35)'
        WHEN Age <= 55 THEN '中年(36-55)'
        ELSE '老年(55+)'
    END AS 年龄段,
    COUNT(*) AS 总人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
GROUP BY 年龄段
ORDER BY MIN(Age);
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-- 第六部分：单变量分析 - 登船港口
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-- 6.1 登船港口缺失情况统计
SELECT
    COUNT(*) AS 总人数,
    COUNT(Embarked) AS 有港口记录,
    COUNT(*) - COUNT(Embarked) AS 港口缺失,
    ROUND(COUNT(Embarked) * 100.0 / COUNT(*), 2) AS 完整率百分比
FROM passengers;

-- 6.2 各港口生还率分析
SELECT
    CASE
        WHEN Embarked IS NULL THEN '未知'
        WHEN Embarked = 'C' THEN '瑟堡(Cherbourg)'
        WHEN Embarked = 'Q' THEN '皇后镇(Queenstown)'
        WHEN Embarked = 'S' THEN '南安普顿(Southampton)'
    END AS 登船港口,
    COUNT(*) AS 总人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
GROUP BY Embarked
ORDER BY 生还率百分比 DESC;


-- ==========================================
-- 第七部分：多变量分析 - 性别 × 舱位
-- ==========================================

-- 7.1 性别与舱位交叉分析
SELECT
    Pclass AS 舱位,
    CASE WHEN Sex = 'male' THEN '男性' ELSE '女性' END AS 性别,
    COUNT(*) AS 总人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
GROUP BY Pclass, Sex
ORDER BY Pclass, 性别 DESC;


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-- 第八部分：多变量分析 - 年龄 × 舱位
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-- 8.1 各舱位儿童与成年人对比
SELECT
    Pclass AS 舱位,
    CASE
        WHEN Age <= 12 THEN '儿童(0-12岁)'
        ELSE '成年人(>12岁)'
    END AS 人群,
    COUNT(*) AS 总人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
WHERE Age IS NOT NULL
GROUP BY Pclass, 人群
ORDER BY Pclass, 人群;


-- ==========================================
-- 第九部分：多变量分析 - 家庭规模
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-- 9.1 家庭规模对生还率的影响
SELECT
    SibSp + Parch + 1 AS FamilySize,
    COUNT(*) AS 总人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
GROUP BY FamilySize
ORDER BY FamilySize;


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-- ==========================================
-- 第十部分：生还者画像验证
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-- 10.1 画像A：精英儿童（1等舱+儿童+有家庭）
SELECT
    '画像A：精英儿童' AS 画像,
    COUNT(*) AS 人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
WHERE Pclass = 1
  AND Age <= 12
  AND (SibSp + Parch + 1) > 1;

-- 10.2 画像B：中产家庭女性（2等舱+女性+中等家庭）
SELECT
    '画像B：中产家庭女性' AS 画像,
    COUNT(*) AS 人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
WHERE Pclass = 2
  AND Sex = 'female'
  AND (SibSp + Parch + 1) BETWEEN 2 AND 4;

-- 10.3 画像C：独行青年男性（3等舱+男性+青年+单人）
SELECT
    '画像C：独行青年男性' AS 画像,
    COUNT(*) AS 人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
WHERE Pclass = 3
  AND Sex = 'male'
  AND Age BETWEEN 19 AND 35
  AND (SibSp + Parch + 1) = 1;

-- 10.4 黄金群体（1等舱+女性+20-40岁+有家庭）
SELECT
    '黄金群体' AS 画像,
    COUNT(*) AS 人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers
WHERE Sex = 'female'
  AND Pclass = 1
  AND Age BETWEEN 20 AND 40
  AND (SibSp + Parch + 1) > 1;

-- 10.5 全体乘客对比基准
SELECT
    '全体乘客' AS 画像,
    COUNT(*) AS 人数,
    SUM(Survived) AS 生还人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率百分比
FROM passengers;


-- ==========================================
-- 第十一部分：快速验证查询
-- ==========================================

-- 11.1 舱位与舱位分布验证（解释港口差异）
SELECT
    Embarked,
    Pclass,
    COUNT(*) AS 人数,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (PARTITION BY Embarked), 2) AS 占比
FROM passengers
WHERE Embarked IS NOT NULL
GROUP BY Embarked, Pclass
ORDER BY Embarked, Pclass;

-- 11.2 儿童性别差异（补充分析）
SELECT
    CASE WHEN Sex = 'male' THEN '男童' ELSE '女童' END AS 性别,
    COUNT(*) AS 人数,
    ROUND(AVG(Survived) * 100, 2) AS 生还率
FROM passengers
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WHERE Age <= 12
GROUP BY Sex;
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